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Neural surrogate priors for Bayesian hierarchical models: what does DeepRV preserve, and what does it distort?

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The work contained in this thesis is my own work unless otherwise stated.

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Abstract

Gaussian process (GP) priors are widely used in hierarchical models for temporal and spatio-temporal data. However, exact GP inference scales as $\mathcal{O}(N^3)$, which can become computationally prohibitive for large national-level datasets. Two approximation methods are compared in this thesis; both reduce costs by transforming the latent GP representation, and neither alters the sampling procedure nor the inferential objective. They are the Hilbert-space approximate GP (HSGP) and DeepRV, which represents the latent field using a finite basis of Laplacian eigenfunctions and uses a neural decoder trained offline on draws from an exact GP respectively.

Four real datasets and one synthetic benchmark will be used in this thesis. The two temporal examples are daily birth counts from the United States (1969–1988, $T = 7,305$) and the Netherlands (1995–2024, $T = 10,958$). For both cases, the latent temporal effect is separated into a slow-moving trend, an annual seasonal component and a weekday effect, with different covariance structures used for each component. The main spatio-temporal examples are measles surveillance data from ECDC (29 countries over 120 months) and Project Tycho in the United States (49 states over 420 months). We include a space-time interaction term for these models. We also use a synthetic panel for which the data generating process is known so we can directly recover the true function to be assessed.

The results separate two properties that are usually reported together. Latent fields are preserved: posterior mean trends correlate at 0.9998 and 0.9999 on the two birth series, and held-out predictive density on ECDC is statistically indistinguishable between methods. The hyperparameter posteriors are not. On the Dutch series the trend lengthscale sits at 0.41 under DeepRV against 0.17 under HSGP, a displacement of three pooled posterior standard deviations, while the seasonal, noise and intercept posteriors, none of which pass through the decoder, agree to four decimal places. A plot of the fitted components does not reveal this. The mechanism is that a variance-normalised decoder makes the lengthscale a pure shape parameter, so that the likelihood constrains it only weakly; the same mechanism leaves a spatial lengthscale on ECDC that the data cannot identify at all. Cost does not follow a single pattern. With every hyperparameter sampled throughout, DeepRV costs the same as HSGP on the US series, 27% less on the Dutch series and six times less on ECDC, so the speed advantage claimed for neural surrogate priors does not transfer uniformly across problems.

A neural surrogate prior is therefore suitable when the latent field or its predictive consequences are the estimand, and unsuitable when the kernel hyperparameters are themselves of scientific interest.

1 Introduction

Measles surveillance data can be represented as counts observed for each region and time period. These counts are often sparse and may show substantial spatial and temporal dependence. Rather than modelling these patterns through only observed covariates, a latent log-risk surface is introduced as it can vary smoothly over space and time. Gaussian process (GP) priors provide a flexible way to model this latent structure while accounting for uncertainty in the estimated risk surface, and have been widely used in spatial and spatio-temporal disease modelling (Diggle et al., 1998; Rasmussen & Williams, 2006).

The main difficulty is computational. Exact GP inference requires Cholesky factorisation of an $N \times N$ covariance matrix with the scales as $\mathcal{O}(N^3)$. This is expensive when N is a few thousand, and becomes increasingly difficult when the datasets get larger. GP approximations make inference more tractable by simplifying the presentation of latent process. However, different approximations could have different impacts on computational performance and posterior inference.

There are many approaches that have been developed to reduce the computational cost of GP models. Inducing-point and sparse methods represent the process using a smaller set of pseudo-inputs (Quiñonero-Candela & Rasmussen, 2005; Snelson & Ghahramani, 2005), while low-rank approaches such as Nyström approximations and random Fourier features (Rahimi & Recht, 2007) provide alternative approximations to the covariance structure. Vecchia methods factorise the joint distribution into a sequence of lower-dimensional conditional distributions, and structured kernel interpolation makes use of regular grid structure. INLA (Rue et al., 2009), by contrast, avoids MCMC and uses a deterministic Laplace-based approximation.

Here we focus on methods that change the representation of the latent function while keeping the sampling procedure and inferential target as unchanged. Two such methods are considered. The first is the Hilbert-space GP approximation (HSGP). It represents the latent GP using a finite set of basis functions. Solin & Särkkä (2020) developed an approximation based on eigenfunctions of the Laplace operator, where the contribution of each basis function is determined by the spectral density of the covariance function. Thanks to Riutort-Mayol et al. (2023), a practical guidance is provided to choose the number of basis functions and the size of the approximation domain. The linear representation of the latent GP in terms of its basis weights helps avoid working directly with the full covariance matrix, so HSGP provides a useful basis-function benchmark against which to compare DeepRV.

The other approach is to learn a neural approximation to the GP prior. PriorVAE (Semenova et al., 2022) trains a variational autoencoder on draws from the prior and

then uses the trained decoder in place of the GP during MCMC. PriorCVAE (Semenova et al., 2023) extends this idea by conditioning the network on the GP hyperparameters, so that these parameters can still be inferred. DeepRV (Navott et al., 2025) instead uses a decoder-only architecture which trained to map latent Gaussian inputs directly to GP realisations. It removes the encoder and the KL term used in VAE training, which enables the network to focus on reconstructing the target GP draws. In each case, part of the computational work is moved to an offline training stage, after which the trained network is used during Bayesian inference.

The DeepRV paper sets benchmarks to compare against INLA, inducing-point GPs, random Fourier features, ADVI, PriorCVAE and a Vecchia approximation, in each case measuring fidelity to full GP MCMC, but the HSGP has not appeared yet. We regard the omission as notable, because HSGP is arguably DeepRV’s closest competitor. Both replace the GP prior with a cheaper representation that slots into an otherwise unchanged MCMC sampler. Both retain the full posterior rather than approximating it, as INLA and ADVI do. HSGP is also the reduced-rank GP a practitioner is most likely to reach for, being built into Stan, `brms` and NumPyro. The comparators used are either approximate-inference schemes that change the inferential target (INLA, ADVI) or sparse approximations that address a different bottleneck (inducing points, RFF, Vecchia). HSGP performs the same task by the same route.

The limitation is clearer in the real-data example. DeepRV is applied to education deprivation across 4,994 London LSOAs, but this experiment compares it only with a GP baseline rather than with another approximation. This leaves a direct comparison with HSGP unresolved. In particular, it is not clear whether the advantages of a learned surrogate are maintained when it is compared with a conventional basis approximation on real data. The applications considered here examine this question in more structured temporal and spatio-temporal models, including count data with a space-time interaction.

Hence we introduce the measles data to which DeepRV has not previously been exposed: a national count data with over-dispersion, a space–time interaction, and GP hyperparameters estimated from the data rather than fixed in advance.

Contributions. We construct a strictly matched benchmark of HSGP against DeepRV. We run it across four real-data applications and one synthetic benchmark, spanning 1,440 to 19,871 observations. Two dailybirth series serve as auxiliary temporal case studies. Two measles panels are the primary spatio-temporal application, using a negative-binomial model with a learned space–time interaction. We evaluate on held-out data rather than in-sample fit, which turns out to reverse the impression given by the fitted

curves. And we isolate the space–time interaction as the single component that separates the methods, by refitting a simple model without it.

2 Data

Four real-data applications and one synthetic benchmark are considered. The two daily birth series are used as simpler temporal case studies. Their one-dimensional structure allows slow and fast trend, seasonal pattern and weekday effect to be examined separately. The two measles surveillance panels provide the main spatio-temporal applications and include a space–time interaction. We also use a synthetic dataset with known latent functions, which allows recovery of the underlying process to be assessed directly. We would start with describing the main features of each dataset before introducing the corresponding models.

2.1 Daily births

United States, 1969–1988. Daily counts of live births, with $T = 7,305$ observations with an average of 9,649 births per day. **Netherlands, 1995–2024.** Daily birth counts from Statistics Netherlands (CBS), with $T = 10,958$ observations and an average of 500 births per day. The two series have similar temporal patterns despite the large difference in their average counts. This difference is relevant to the likelihood approximation discussed in §3.3.

Figure 1 shows three clear patterns in both series. The long-term trend varies between 89% and 111% of the series mean in the US and 113–91% in the Netherlands. An annual cycle of around $\pm 7\%$ peaks in early autumn in both countries. A weekday effect places weekends 10–18% below weekdays, which is consistent with the scheduling of induced and planned deliveries.

The middle panels suggests clear deviations from the smooth trend on specific calendar days. Birth counts fall sharply on major holidays and are followed by increases of around 8–11% on the next few days. Christmas falls to about 80% of the mean in the US, while New Year’s Day falls to about 72% in the Netherlands. The relevant holidays also differ between the two countries, so these short-term effects are country-specific.

Two daily birth series: the same three structures, at different scales and with different holidays

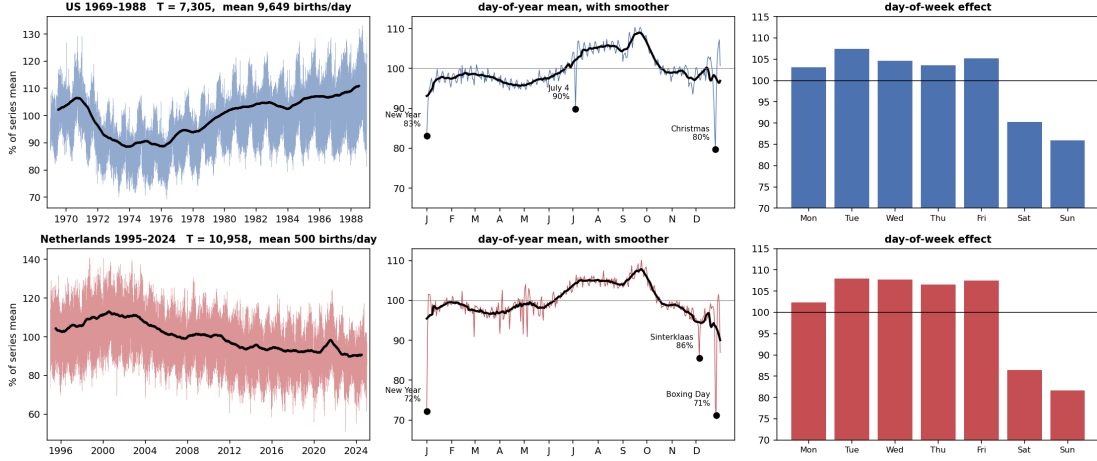


Figure 1: The two daily birth series. Left: raw daily counts as a percentage of the series mean, with a one-year rolling mean. Middle: day-of-year mean (thin) with a 15-day smoother (thick); marked points are holidays the smoother cannot reach. Right: weekday effect. The same three structures appear in both countries, at different scales and with different national holidays.

2.2 Synthetic panel

When dealing with real data, the true latent field is unknown, so recovery cannot be assessed directly. We therefore would first consider a synthetic spatio-temporal panel with known data generating process. The dataset contains twenty regions with fixed coordinates in $[0, 1]^2$, log-normal population sizes and observations over 72 months. Counts are generated from a Poisson model with a baseline rate of 1.2 per 100,000. The latent structure follows that used for the measles applications, including a spatial effect, a temporal trend, a separable space-time interaction and a fixed seasonal component. The latent field is generated at full rank using exact Cholesky factorisation. This ensures that the simulated data do not favour either approximation. The interaction is generated as $w = L_{ws} Z L_{wt}^\top$, where Z has independent standard normal entries, and is then centred across both space and time, which prevents the interaction from absorbing the spatial or temporal main effects which may identify the separate components of the model. The true hyperparameters are $(\alpha_g, \ell_g) = (0.30, 0.55)$, $(\alpha_q, \ell_q) = (0.20, 18)$ months, and $(\alpha_w, \ell_{ws}, \ell_{wt}) = (0.18, 0.15, 6)$. The interaction lengthscale $\ell_{ws} = 0.15$ can be represented well with a relatively small basis. The synthetic example hence provides a setting in which both HSGP and DeepRV should be able to recover the main structure of the latent field. It serves as a reference point before moving to the complex real-data applications. The simulation is repeated for three seeds and coverage is evaluated across

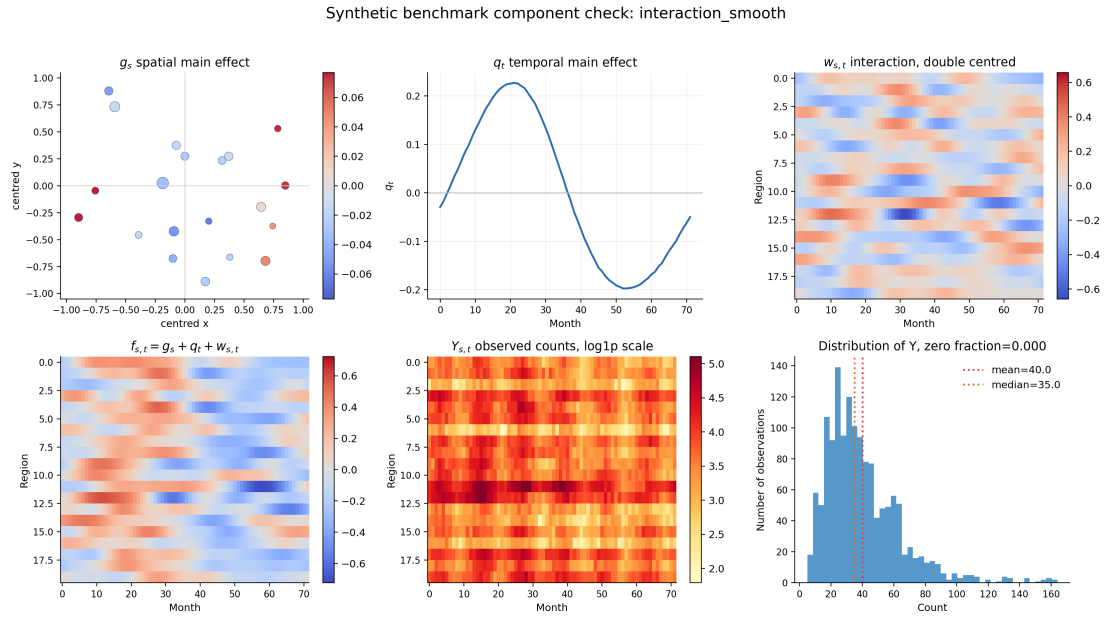


Figure 2: One realisation of the synthetic panel, decomposed. Top: the spatial main effect over region coordinates, the temporal main effect, and the doubly centred space–time interaction. Bottom: the summed latent field $f_{s,t} = g_s + q_t + w_{s,t}$, the observed counts on a $\log(1 + y)$ scale, and the count distribution. The interaction dominates the latent field, and the counts contain no zeros.

several simulated realisations.

Figure 2 shows one simulated realisation together with its individual components. The space–time interaction has a larger amplitude than either main effect and accounts for much of the variation in the combined latent field. After centring across both dimensions, there is no persistent spatial or temporal offset in the interaction. Unlike the surveillance datasets, the simulated counts contain no zeros and show little evidence of the sparsity encountered in the real data.

2.3 Measles surveillance

The two real panels use the same broad additive structure as the synthetic example, but present more challenging features in the observed data. Figure 3 shows the two datasets using the same summaries as for the synthetic panel.

ECDC. We use confirmed measles cases reported to the European Centre for Disease Prevention and Control, aggregated by country and month (ECDC, 2026). The analysis covers 2010–2019, ending before the COVID-19 pandemic, during which reporting was disrupted. Including this period would introduce a structural change that is unlikely to be captured well by a smooth GP. We remove the pan-European aggregate row and one country with no reported cases during the study period, leaving 29 countries observed over 120 months. Exposure is represented using a time-varying population offset based on Eurostat population estimates for 1 January of each year (online data code `demo_pjan`, total population), retrieved through the Eurostat REST API (Eurostat, 2026). This corresponds to the denominator used by ECDC when reporting incidence rates. Country locations are represented by centroid longitude and latitude, with a cosine-latitude correction applied to longitude. This increases the correlation between Euclidean and great-circle distances from 0.94 to 0.98.

Project Tycho. We use digitised US notifiable-disease records from Project Tycho (van Panhuis et al., 2013). The analysis covers the pre-vaccine period from 1928 to 1962, with weekly records summed to months, giving 49 states observed over 420 months. This period contains a strong epidemic signal while avoiding the structural change associated with widespread measles vaccination.

Figure 3 shows why these two panels demand a different model from the birth series, and why they differ from each other. Three features are relevant.

The two datasets differ substantially in sparsity and dispersion. In the ECDC panel, 54% of country-month observations are zero and the overall variance-to-mean ratio is 1,134. In Project Tycho, only 0.2% of observations are zero, but the variance-to-mean ratio is much larger at 8,382. Given the high variance-to-mean ratio, a Poisson model is not suited for these two data sets. There are significant number of zero-count months, particularly between outbreaks. The Tycho data in contrast contains much larger epidemic peaks. Hence negative binomial likelihood is used as it has the dispersion parameter estimated from the data in (15).

Outbreaks are asynchronous across regions, and this is the structure the interaction term exists to represent. In the ECDC data, major outbreaks occur in different years, with Bulgaria experiencing high case counts early in the decade and Italy and Romania later in the period, and in between the case number of most countries is near zero for years at a time. Plot in panel (b) suggests that these outbreaks appear as localised high-count regions rather than vertical bands shared across countries. For US measles data, though

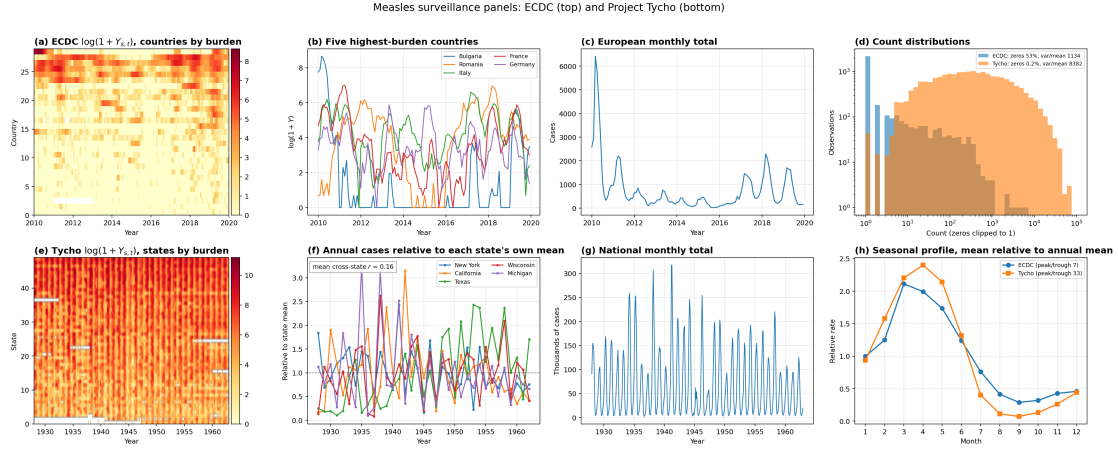


Figure 3: The two measles panels. (a) ECDC counts by country and month, ordered by total burden. (b) Monthly trajectories for the five highest-burden countries, showing outbreaks at different times. (c) European monthly total, with several large epidemic peaks. (d) Count distributions for ECDC and Tycho, illustrating their contrasting sparsity and dispersion. (e) Tycho counts by state and month, showing strong recurrent epidemic bands. (f) Annual cases for selected states relative to each state’s mean, with substantial variation across states and years. (g) National monthly total, showing the pronounced pre-vaccine epidemic cycle. (h) Seasonal profiles for the two datasets, with broadly similar timing but stronger seasonality in Tycho.

the peak of each state roughly resonates with the national level which can be observed from panel (f) and (g), individual state still has their own variation in timing. A model with only a shared temporal trend q_t cannot capture these country-specific changes in outbreak timing, which is visibly inconsistent with these data. The effect of removing this interaction is examined in §5.6.

Seasonality is strong and similar in shape, but differs in amplitude. Both series peak in spring and trough in September, but the Tycho peak-to-trough ratio is 34 against ECDC’s 7. The seasonal component needs to be estimated rather than fixed, as its amplitude cannot be transferred between datasets.

3 Background

A zero-mean Gaussian process (GP) is defined such that, for any finite set of inputs $x_1, \dots, x_N$, the corresponding function values $\mathbf{f} = (f(x_1), \dots, f(x_N))$ follow a multivariate normal distribution,

$\mathbf{f} \sim \mathcal{N}(\mathbf{0}, \mathbf{K})$, where $K_{ij} = k(x_i, x_j)$. Throughout this thesis, we use the squared-exponential kernel,

$$k(x, x') = \alpha^2 \exp\left(-\frac{\|x - x'\|^2}{2\ell^2}\right), \quad (1)$$

where α is the marginal standard deviation and ℓ is the lengthscale. The corresponding correlation decreases as

$$\text{Corr}\{f(x), f(x')\} = \exp\left(-\frac{\|x - x'\|^2}{2\ell^2}\right),$$

so smaller values of ℓ allow the process to vary over shorter spatial or temporal scales, while larger values imply more persistent dependence over distance.

Exact GP requires a Cholesky factorisation of the covariance matrix,

$\mathbf{K} = \mathbf{L}\mathbf{L}^\top$. In column j of the factorisation, approximately $N - j$ entries must be computed, with each involving an inner product of length proportional to j , so the computational cost scales as

$$\sum_{j=1}^N (N - j)j = \frac{N(N + 1)(N - 1)}{6} = \mathcal{O}(N^3).$$

Such operations gives a leading-order cost of approximately $\frac{1}{3}N^3$ for the Cholesky factorisation. When covariance hyperparameters are inferred within MCMC, the covariance matrix changes during sampling, so repeated factorisations can be very computationally expensive. Hence we introduce methods that can reduce such cost.

3.1 The Hilbert-space approximation

The HSGP approximation starts from the eigenvalue problem for the negative Laplace operator on a bounded domain $\Omega = [-L, L]^d$, with Dirichlet boundary conditions,

$$-\nabla^2 \phi_j(x) = \lambda_j \phi_j(x) \quad \text{on } \Omega, \quad \phi_j(x) = 0 \quad \text{on } \partial\Omega. \quad (2)$$

Here, ϕ_j is the j th eigenfunction and λ_j is the corresponding eigenvalue. These eigenfunctions form an orthogonal basis on the bounded domain. In one dimension, we have

the closed-form expressions

$$\phi_j(x) = \frac{1}{\sqrt{L}} \sin\left(\frac{\pi j(x+L)}{2L}\right), \quad \lambda_j = \left(\frac{\pi j}{2L}\right)^2, \quad j = 1, 2, \dots \quad (3)$$

The eigenfunctions vanish at the boundaries $-L$ and L . The quantity $\sqrt{\lambda_j} = \pi j/(2L)$ is the spatial frequency of the j th basis function, so larger values of j correspond to increasingly rapid variation over the domain. In d dimensions, the eigenfunctions are formed by products of the one-dimensional eigenfunctions. For a multi-index $\mathbf{j} = (j_1, \dots, j_d)$, the corresponding eigenvalue is

$$\lambda_{\mathbf{j}} = \sum_{r=1}^d \left(\frac{\pi j_r}{2L}\right)^2.$$

The connection between the Laplacian eigenbasis and a stationary GP comes from the spectral representation of the covariance function. By Bochner's theorem, a stationary kernel can be represented as the Fourier transform of a non-negative spectral density $S_{\theta}(\omega)$. In the Fourier domain, the negative Laplace operator acts as multiplication by the squared frequency,

$$-\nabla^2 \longleftrightarrow \|\omega\|^2.$$

Hence a Laplacian eigenvalue λ_j corresponds to a frequency with magnitude $\sqrt{\lambda_j}$. This explains why the spectral density is evaluated at $S_{\theta}(\sqrt{\lambda_j})$ in the HSGP construction.

On a sufficiently large bounded domain, the Laplacian eigenfunctions approximately diagonalise the covariance operator (Solin & Särkkä, 2020), giving

$$k(x, x') \approx \sum_j S_{\theta}(\sqrt{\lambda_j}) \phi_j(x) \phi_j(x').$$

The basis functions ϕ_j determine the spatial shape of each component, while $S_{\theta}(\sqrt{\lambda_j})$ determines how much prior variance the GP assigns to the corresponding frequency.

Truncating the expansion after m^* basis functions gives the finite-dimensional approximation

$$f(x) \approx \sum_{j=1}^{m^*} \sqrt{S_{\theta}(\sqrt{\lambda_j})} \phi_j(x) \beta_j, \quad \beta_j \sim \mathcal{N}(0, 1). \quad (4)$$

The coefficients β_j are independent standard-normal variables, while the factor $\sqrt{S_{\theta}(\sqrt{\lambda_j})}$ scales each basis function according to the covariance kernel. The Laplacian basis depends only on the domain and boundary conditions, rather than on the GP hyperparameters, so changing a hyperparameter such as the lengthscale ℓ does not require a

new $N \times N$ Cholesky factorisation. The latent field is also linear in the standard-normal coefficients β_j , giving a convenient non-centred parameterisation.

Computationally, this replaces the $\mathcal{O}(N^3)$ cost of repeated dense covariance factorisations with a linear basis expansion. For a univariate HSGP with m basis functions, evaluating the log posterior scales as

$$\mathcal{O}(Nm + m),$$

where N is the number of observations. In d dimensions, the total number of basis functions is $m^* = \prod_{r=1}^d m_r$, giving a corresponding cost of

$$\mathcal{O}(Nm^* + m^*).$$

When $m^* \ll N$, this is substantially cheaper than the $\mathcal{O}(N^3)$ scaling of an exact GP, although the advantage decreases as the number of basis functions grows with dimension.

The quality of the approximation depends on both the size of the extended domain and the number of retained basis functions. [Riutort-Mayol et al. \(2023\)](#) provide practical recommendations for the choice of quantities. For the squared-exponential kernel,

$$m \geq 1.75 \frac{c}{\ell/S}, \quad c = \frac{L}{S} \geq \max\{3.2(\ell/S), 1.2\}, \quad (5)$$

where S is the half-range of the observed input domain and $L = cS$ is the half-range of the enlarged HSGP domain. The first condition controls the number of basis functions, while the second controls how far the artificial boundaries are placed from the observed data.

Since truncating the expansion at m also limits the highest frequency that can be represented, shorter lengthscales generally require larger m . In multiple dimensions the basis is constructed as a tensor product, so the total number of basis functions is

$$m^* = \prod_{r=1}^d m_r.$$

The basis size therefore grows multiplicatively with dimension. For example, using m basis functions in each of two dimensions gives m^2 terms, while three dimensions give m^3 . This growth becomes important for higher-dimensional interactions.

3.2 DeepRV

DeepRV (Navott et al., 2025) replaces repeated GP covariance calculations during inference with a neural decoder which is trained offline on exact GP realisations. At inference, the latent field is generated by

$$f = \alpha g_\phi(\mathbf{z}, \log \ell), \quad \mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_r), \quad (6)$$

where g_ϕ is a deterministic decoder with trained parameters ϕ . The lengthscale ℓ is supplied to the decoder as a conditioning input, while the amplitude α is applied outside the network. The decoder is trained on variance-normalised GP realisations and is frozen before posterior inference.

A forward pass through the decoder can be written as a sequence of transformations. Let

$$\mathbf{h}^{(0)} = [\mathbf{z}, \log \ell]$$

denote the input to the network. For a decoder with K layers,

$$\mathbf{h}^{(k)} = \sigma_k(\mathbf{W}_k \mathbf{h}^{(k-1)} + \mathbf{b}_k), \quad k = 1, \dots, K, \quad (7)$$

where $\mathbf{W}_k$ and $\mathbf{b}_k$ are the weights and biases of layer k , and σ_k denotes its activation function. The final output $\mathbf{h}^{(K)} = g_\phi(\mathbf{z}, \log \ell)$ gives the variance-normalised latent field, which is then multiplied by α in (6). Once training is complete, the matrices $\mathbf{W}_k$ and vectors $\mathbf{b}_k$ would remain fixed during MCMC.

The inference uses matrix-vector operations through a fixed network rather than repeated factorisation of an $N \times N$ covariance matrix. If layer k has input width h_{k-1} and output width h_k , multiplication by $\mathbf{W}_k$ requires work proportional to $h_{k-1}h_k$. The cost of one forward pass therefore scales as

$$C_{\text{DeepRV}} = \mathcal{O}\left(\sum_{k=1}^K h_{k-1}h_k\right). \quad (8)$$

For the architectures considered in Navott et al. (2025), this gives approximately $\mathcal{O}(N^2)$ scaling with the field size, although the exact cost depends on the decoder architecture. This replaces the $\mathcal{O}(N^3)$ covariance factorisation required by an exact GP.

In our implementation, the latent dimension r is smaller than the dimension of the output field. For example, the ECDC interaction uses 387 latent variables to generate a field with 3,450 entries. This reduces the dimension explored by NUTS, but also introduces an additional approximation because the decoder can only generate fields represented

by the map g_ϕ . The computational gain must therefore be considered together with the fidelity of the learned prior.

We assess this fidelity using the variance ratio

$$\rho^2 = \frac{\text{Var}(\hat{f})}{\text{Var}(f)}, \quad (9)$$

where $\hat{f}$ denotes a field generated by the trained decoder and f denotes a draw from the target GP prior. Values of ρ^2 below one indicate that the decoder reproduces less marginal variation than the target process. Since matching the variance alone does not guarantee that the full covariance structure is reproduced, this diagnostic is considered together with the other measures of prior and posterior fidelity reported later.

Unlike HSGP, there is no analytic rule analogous to (5) that determines the latent dimension required for a given approximation accuracy. In our implementation, r is chosen using a principal-component analysis of prior draws, retaining enough components to represent most of the prior variation. This provides a practical dimension-reduction rule, but remains a heuristic rather than a theoretical approximation guarantee.

3.3 The births model

The observation model is

$$y_t \mid \eta_t, \sigma \sim \mathcal{N}(\eta_t, \sigma), \quad \eta_t = \beta_0 + f(t) + h(d_t) + \beta_{w_t}. \quad (10)$$

The model consists of a long-term trend, an annual effect and a weekday effect.

Before evaluating the covariance function, time is linearly mapped to $[-1, 1]$ using

$$t_{\text{std}} = 2 \frac{t - t_{\min}}{t_{\max} - t_{\min}} - 1.$$

The lengthscale of the long-term component is measured relative to the total duration of the observed series.

The long-term trend f is assigned a squared-exponential GP prior,

$$f \sim \text{GP}(0, k_f), \quad k_f(t, t') = \alpha_f^2 \exp\left(-\frac{(t - t')^2}{2\rho_f^2}\right), \quad (11)$$

with

$$\alpha_f \sim \text{HalfNormal}(1), \quad \rho_f \sim \text{InverseGamma}(4.047, 2.530).$$

The prior for ρ_f has a 98% interval of $[0.25, 3.00]$ on the standardised time scale. The full standardised time domain has width 2, where smaller values of ρ_f allow variation over a shorter fraction of the observation period and larger values imply dependence over a wider part of the series so the prior favours gradual long-term variation by assigning little probability to extremely short lengthscales.

The annual component h is assigned a periodic GP prior,

$$h \sim \text{GP}(0, k_h), \quad k_h(d, d') = \alpha_h^2 \exp\left(-\frac{2 \sin^2(\pi|d - d'|/P)}{\rho_h^2}\right), \quad P = 365.25, \quad (12)$$

with

$$\alpha_h \sim \text{HalfNormal}(1), \quad \rho_h \sim \text{InverseGamma}(3.934, 1.193).$$

The 98% prior interval for ρ_h is $[0.12, 1.50]$. The periodic covariance treats the end and beginning of the calendar year as adjacent, avoiding an artificial discontinuity between late December and early January.

The weekday effect is represented by

$$\boldsymbol{\beta} = (\beta_1, \dots, \beta_7) \sim \text{ZeroSumNormal}(1), \quad \sum_{w=1}^7 \beta_w = 0.$$

The zero-sum constraint separates the weekday effects from the intercept: β_0 represents the overall level, while β_w represents the deviation associated with weekday w .

The comparison between HSGP and DeepRV changes only the representation of the long-term trend f . Under HSGP, f is represented using the basis expansion described in Section 3.1. Under DeepRV,

$$\mathbf{u} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_r), \quad f = D_\theta(\mathbf{u}, \alpha_f, \rho_f), \quad (13)$$

where D_θ is trained to reproduce draws from the same GP prior for f .

3.4 The disease model

Let $s \in \{1, \dots, S\}$ be the index of regions, corresponding to countries in the ECDC data and states in Project Tycho, and let $t \in \{1, \dots, T\}$ be the index of months. The observed count in region s and month t is denoted by $y_{s,t}$, and $m(t) \in \{1, \dots, 12\}$ denotes the calendar month. Both datasets show obvious variation compared with the expected under a simple Poisson model, so we use a negative binomial likelihood in its

mean–dispersion parameterisation,

$$y_{s,t} \sim \text{NegBin2}(\exp(\eta_{s,t}), \kappa), \quad (14)$$

$$\eta_{s,t} = \log E_{s,t} + \beta_0 + h_{m(t)} + \underbrace{g_s + q_t + w_{s,t}}_{\text{latent field } f_{s,t}}, \quad (15)$$

where $E_{s,t}$ is the population exposure (which is typically the population size) and β_0 is the intercept. Under this parameterisation,

$$\mathbb{E}(y_{s,t}) = \mu_{s,t}, \quad \text{Var}(y_{s,t}) = \mu_{s,t} + \frac{\mu_{s,t}^2}{\kappa}, \quad \mu_{s,t} = \exp(\eta_{s,t}),$$

so finite κ allows additional variation relative to a Poisson likelihood.

The latent structure has four components. The spatial main effect is $g \sim \mathcal{GP}(0, k_g)$ over region coordinates. The temporal trend is $q \sim \mathcal{GP}(0, k_q)$ over month. The seasonal term h is modelled using a 12-month periodic covariance. The space–time interaction is

$$w \sim \mathcal{GP}(0, k_{ws} \otimes k_{wt}).$$

The kernels k_g , k_q , k_{ws} and k_{wt} use the squared-exponential form in (1), with separate amplitudes and lengthscales that are inferred from the data. The spatial, temporal and interaction components are centred so that the additive decomposition remains identifiable.

For the dispersion term we use

$$\frac{1}{\sqrt{\kappa}} \sim \text{HalfNormal}(1),$$

following the Stan prior recommendations (Stan Development Team, 2026) and the penalised-complexity argument of Simpson et al. (2017). Since the prior has highest density at near 0, the prior tends to shrink towards the Poisson model but still allows additional dispersion.

3.5 Containment priors and additive separation

Inference for a GP lengthscale is degenerate at both ends (Betancourt, 2020). When the lengthscale is below the shortest informative separation between observations, further reductions cause little distinguishable change in the likelihood. When the lengthscale greatly exceeds the input domain, the function is nearly constant over the observed range, so the likelihood provides little information about its exact value.

We control these regions using an inverse-gamma containment prior. Let $\rho_{\min}$ denote the minimum informative distance, taken here as the smallest nearest-neighbour distance in the input, and let $\rho_{\max}$ denote the span of the input domain. For $\rho \sim \text{InverseGamma}(a, b)$, the parameters are chosen to satisfy

$$\Pr(\rho < \rho_{\min}) = 0.01, \quad \Pr(\rho > \rho_{\max}) = 0.01. \quad (16)$$

The parameters (a, b) are obtained numerically by solving these two equations. This places 98% of the prior mass between $\rho_{\min}$ and $\rho_{\max}$. Each lengthscale prior is constructed using (16) and reported in Table 2.

When several latent components enter the model additively, large overlap of lengthscale priors should be avoided as different components may explain similar patterns in the data, leading to weak identifiability. (Betancourt, 2020) proposes that priors should only have negligible overlap.

4 Methodology

4.1 Design of the comparison

Here 5 settings are summarised in (Table 1). The two birth series are 1-dimensional temporal case studies. The synthetic panel is the only setting with ground truth, so it is the only place where recovery error can be measured directly. The two measles panels are the primary applications, about which we have no truth, so we judge them on out-of-sample prediction and calibration.

Table 1: The five benchmark settings: four real-data applications and one synthetic benchmark.

Dataset	Role	Resolution	n	Likelihood	Truth
US births, 1969–1988	auxiliary	daily	7,305	Gaussian	no
NL births, 1995–2024	auxiliary	daily	10,958	Gaussian	no
Synthetic	benchmark	monthly, 20×72	1,440	Poisson	yes
ECDC (Europe)	primary	monthly, 29×120	3,450	negative binomial	no
Tycho (US, 1928–62)	primary	monthly, 49×420	19,871	negative binomial	no

Auxiliary: one-dimensional temporal case studies. Primary: spatio-temporal applications with a learned space–time interaction.

The HSGP and DeepRV fits were constructed to use the same model specification wherever possible, including the squared-exponential covariance, the exp-sine-squared seasonal component evaluated through an exact 12×12 Cholesky factorisation, the inverse-

gamma lengthscale priors, the likelihood and the data window, with the main difference being the representation of the latent GP. This includes the squared-exponential covariance, the exp-sine-squared seasonal component evaluated through an exact 12×12 Cholesky factorisation, the inverse-gamma lengthscale priors, the likelihood and the data window. NUTS settings uses four chains, target acceptance probabilities between 0.9 and 0.95, and maximum tree depths between 10 and 12.

4.2 Priors

The lengthscale priors are constructed separately for each dataset using the containment procedure described above. Table 2 reports the resulting priors for ECDC, while the same procedure is applied to the other datasets. We measure the overlap between the spatial main-effect prior and the interaction-space prior at 1.9%, and between the trend and interaction-time priors at 3.0%. Both sit within the range Betancourt’s separation requirement tolerates.

Table 2: Prior specification for the ECDC fits. Spatial half-range $S = 0.61$, minimum nearest-neighbour distance 0.083, $T = 120$ months.

Parameter	Prior	98% interval / role
ℓ_g (spatial)	InvGamma(6.11, 3.32)	[0.25, 1.80]
ℓ_{ws} (interaction, space)	InvGamma(5.37, 0.36)	[0.03, 0.25]
ℓ_q (trend)	InvGamma(6.57, 167.4)	[12, 80] months
ℓ_{wt} (interaction, time)	InvGamma(6.27, 27.0)	[2, 14] months
$\alpha_g, \alpha_q, \alpha_w$	HalfNormal(1)	amplitudes
σ_h	HalfNormal(0.45)	seasonal amplitude
$1/\sqrt{\kappa}$	HalfNormal(1)	over-dispersion
β_0	$\mathcal{N}(\hat{\beta}_0, 3^2)$	log baseline rate
$\beta, \mathbf{z}$	$\mathcal{N}(0, 1)$	non-centred weights

4.3 Sizing the basis and the latent

For the lower-dimensional main effects, the criterion in (5) is easily satisfied. The dimensions of the latent fields and observed data differ slightly because of missing observations. We define the latent field on the complete region–time grid, giving 3,480 cells for ECDC and 20,580 for Tycho, while the likelihood is evaluated only at observed cells ($n = 3,450$ and $n = 19,871$, respectively). The decoder generates the full latent grid, with missing cells omitted only from the likelihood evaluation. These locations remain part of the spatial and temporal correlation structure. The corresponding missing-data proportions are 0.9% for ECDC and 3.4% for Tycho.

The space–time interaction places a stronger requirement on the HSGP basis. Table 3 applies the same diagnostic to each component of the fitted ECDC model. The temporal part of the interaction satisfies the criterion, whereas the spatial part does not. Its fitted spatial lengthscale of 0.078 implies a requirement of approximately 22 basis functions per spatial dimension, compared with the eight currently used. Applying the criterion in full would therefore require

$$22^2 \times 13 \approx 6,300$$

interaction basis functions. This is larger than the number of cells in the ECDC field and would remove much of the computational advantage of using a reduced-rank representation.

We therefore limit the size of the interaction basis and report the resulting shortfall using the same diagnostic. The chosen basis size is sufficient to cover the estimated effective rank of the interaction, with a participation ratio of approximately 360. The DeepRV interaction latent dimension is chosen using the same effective-rank criterion. Effective-rank-based sizing is not the standard HSGP basis-selection procedure, but a practical compromise when the criterion in (5) would otherwise require an impractically large tensor-product basis.

Table 3: Riutort-Mayol diagnostic applied to the fitted ECDC HSGP model.

Component	fitted ℓ	ℓ/S	m required	m used	verdict
spatial g	0.53	0.87	5	8	satisfied
trend q	30.3	0.51	4	35	satisfied
interaction (time)	10.5	0.18	13	15	satisfied
interaction (space)	0.078	0.13	22	8	under-resolved

4.4 Training the DeepRV decoders

Each additive component is assigned a separate decoder, conditioned on the lengthscale parameters of its kernel. The interaction decoder takes $\log \ell_{ws}$ and $\log \ell_{wt}$ as inputs, while each main-effect decoder is conditioned on a single lengthscale. Training fields are generated from the corresponding exact GP by Cholesky factorisation, with lengthscales drawn from the same inverse-gamma priors used during inference. Each draw is then centred and scaled to unit variance, allowing the amplitude parameter to be applied outside the decoder at inference. Interaction fields are additionally double-centred to impose the same identifiability constraint used in the fitted model.

The latent dimension r is selected by applying principal-component analysis to the training draws and retaining sufficient components to capture the prescribed proportion of

prior variation. We use this as a practical sizing criterion for DeepRV, since there is no direct analogue of the HSGP basis-selection rule in (5). The interaction requires a substantially larger latent dimension ($r = 360$) than the two main effects ($r = 15$ and $r = 12$). It also has the lowest reconstruction fidelity, with $\rho^2 = 0.847$, compared with 0.969 and 1.002 for the main effects. These diagnostics indicate that the interaction is the most difficult of the three components to approximate.

Table 4 reports the decoder configuration used for the ECDC fits. The synthetic and Tycho decoders are constructed using the same procedure, with latent dimensions determined separately from their respective training draws. Further details of the architecture, optimiser and training budget are given in Appendix 1.1. Decoder fidelity is evaluated using 1,000 independently generated GP fields with random seeds not used during training. These evaluation draws are sampled directly from the target prior rather than obtained by splitting the training set.

Training all three decoders on CPU takes less than one minute. This cost is incurred only once, after which the decoder parameters are fixed and reused across MCMC chains, simulation seeds and model variants. Relative to the repeated sampling cost, the offline training overhead is therefore small.

Table 4: DeepRV decoder configuration for the ECDC fits. Fidelity is measured on 1,000 independently drawn prior fields; ρ^2 is the ratio of reconstructed to true field variance.

	spatial g	trend q	interaction w
output dimension	29	120	3,480
latent dimension r	15	12	360
conditioning inputs	$\log \ell_g$	$\log \ell_q$	$\log \ell_{ws}, \log \ell_{wt}$
prior RMSE	0.039	0.040	0.354
fidelity ρ^2	0.969	1.002	0.847

4.5 Inference and evaluation

We fit all models with NUTS in NumPyro (Phan et al., 2019). We report:

- **Convergence:** rank-normalised split- $\hat{R}$ (target < 1.01 , Vehtari et al., 2021), minimum effective sample size, and divergent transitions.
- **Calibration:** 95% posterior-predictive coverage and the randomised probability integral transform, which is uniform under correct calibration. The randomisation (Czado et al., 2009) is necessary because the counts are discrete.

- **Predictive accuracy:** WAIC (Vehtari et al., 2017) and, for ECDC, a genuine held-out experiment described below.
- **Structure:** the variance decomposition $\text{var}(g) : \text{var}(q) : \text{var}(w)$, which unlike the raw amplitude hyperparameters is comparable across methods. This matters: HSGP’s α_w multiplies a basis expansion while DeepRV’s multiplies a variance-normalised decoder output, so the two are on different scales by construction.
- **Recovery** (synthetic only): RMSE of the posterior mean latent field against the truth, and coverage of the truth by the 95% credible band.

Held-out cross-validation. In-sample fit rewards flexibility, and WAIC and PSIS-LOO are approximations that can fail when a model is close to saturated. To get a clean answer we withheld a random 15% of the ECDC cells (518 of 3,450) from the likelihood entirely, using a shared mask across all three methods. We refitted each model on the remaining 85% and scored predictions on the cells no model had seen. This is slower than LOO, but it depends on no approximation. We weight this comparison most heavily.

The monthly Tycho HSGP fit needed sampler adaptation before it would mix at all; the sequence is in Appendix 1.3. We removed no basis functions to flatter the diagnostic.

5 Results

5.1 US births: a clean substitution

The US births data provide the simplest setting considered here: a one-dimensional series with three well-separated components. The two methods give nearly identical results (Table 5). Their posterior mean trends have a correlation of 0.9998. Predictive RMSE is 0.0389 for HSGP and 0.0395 for DeepRV, with corresponding coverage of 0.944 and 0.949. Runtime is also similar, at 84 and 90 seconds. Both models converge without divergent transitions, and their fitted decompositions are visually very similar (Figure 14, Appendix). In this relatively simple structured setting, replacing the HSGP representation with the decoder has little effect on either inference or computational cost.

One feature of the fit matters for how the birth series should be read. Several calendar days near the year boundary sit far below the fitted seasonal curve. These are the holiday dips of Figure 1.

Table 5: US daily births, 1969–1988. HSGP marginalises the trend weights; DeepRV samples a 20-dimensional latent through a decoder conditioned on them. All hyperparameters are sampled under both methods. Coverage targets 0.90.

	HSGP	DeepRV
decoder training time (s)	—	≈ 7
NUTS runtime (s)	≈ 84	≈ 90
$\max \hat{R}$	1.002	1.003
divergences	0	0
RMSE (log scale)	0.0389	0.0395
90% predictive coverage	0.944	0.949
trend correlation with HSGP	(reference)	0.9998

5.2 Netherlands births: fields agree, hyperparameters do not

The Dutch series is longer ($T = 10,958$). Every hyperparameter is again sampled under both methods, as in §5.1.

The main difference is computational. HSGP requires 661 seconds, compared with 483 seconds for DeepRV, which means there is the 27% of speedup in runtime of DeepRV. For sampling efficiency, ESS/s increases from 0.79 for HSGP to 2.09 for DeepRV. However, the recovered latent field remains comparable between the two methods. This contrasts with the smaller US births analysis, where the runtimes were similar.

The fitted decompositions remain similar (Figure 13, appendix), which shows the same slow trend with the same inflections, annual cycle, and weekday effects. The correlation between the posterior mean linear predictors is 0.9999 and the mean absolute difference is 0.0010 on the log scale. The component plots suggest that the two methods would be interchangeable here.

However, the hyperparameter posteriors do not agree (Figure 4, Table 6). The trend lengthscale, which passes through the decoder, has posterior mean 0.169 under HSGP and 0.410 under DeepRV, a factor of 2.4 and a 1-Wasserstein distance of 3.03 pooled posterior standard deviations. The trend amplitude shifts by 2.05 pooled standard deviations in the same direction. A longer fitted lengthscale gives a smoother field, and a larger amplitude restores the variance lost to that smoothing. Every hyperparameter that does not pass through the decoder agrees. The seasonal lengthscale and amplitude, the observation noise and the intercept differ by 0.02 to 0.10 pooled standard deviations, which is within Monte Carlo error.

Table 6: Netherlands births: hyperparameter posteriors, HSGP versus full-prior DeepRV. The first two rows pass through the decoder; the remainder do not.

Parameter	HSGP	DeepRV	W_1	W_1/sd	ratio
ρ_f trend lengthscale	0.169 ± 0.024	0.410 ± 0.110	0.2414	3.03	2.43
α_f trend amplitude	0.098 ± 0.026	0.173 ± 0.044	0.0746	2.05	1.76
ρ_h seasonal lengthscale	0.3082 ± 0.0287	0.3066 ± 0.0308	0.0021	0.07	0.99
α_h seasonal amplitude	0.0429 ± 0.0100	0.0428 ± 0.0103	0.0004	0.04	1.00
σ observation noise	0.0630 ± 0.0004	0.0630 ± 0.0004	0.0000	0.10	1.00
β_0 intercept	-0.0108 ± 0.0006	-0.0108 ± 0.0006	0.0000	0.02	1.00

Netherlands daily births: hyperparameter posteriors, HSGP vs full-prior DeepRV
(both models sample every hyperparameter; dashed lines are posterior means)

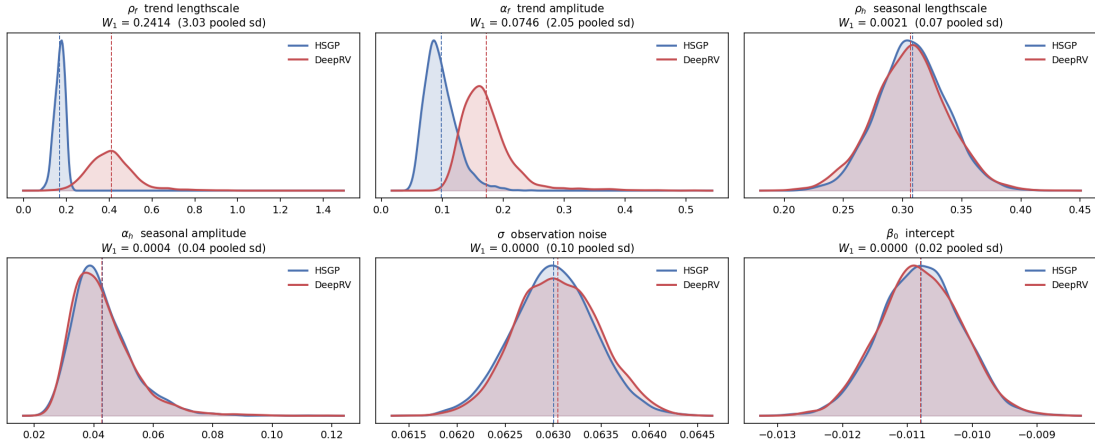


Figure 4: Netherlands births: hyperparameter posterior densities, HSGP (blue) against DeepRV (red)

5.3 Synthetic data, where the truth is known

Figure 5 shows the latent-field recovery of data, which shows both methods recover the main spatial-temporal structure, including the broad positive regions around the middle states and the changes in intensity over time. HSGP gives a slightly lower RMSE than DeepRV, 0.059 compared with 0.075, averaged over three seeds (Table 7). The difference is more visible in interval coverage. HSGP achieves 94.6% coverage for the nominal 95% posterior intervals, while DeepRV achieves 85.7%. The latter still covers the true field at most locations, but its intervals are less well calibrated than those from HSGP. DeepRV is approximately $1.4\times$ faster.

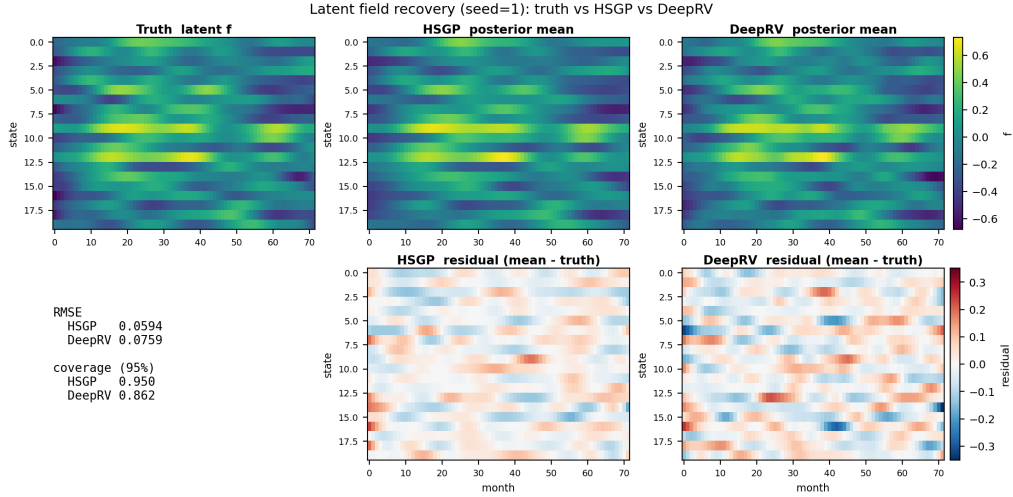


Figure 5: Synthetic latent-field recovery. Top row: truth and the two posterior means. Bottom row: residuals against truth. Both methods reproduce the field; DeepRV’s residuals are modestly larger.

Table 7: Synthetic benchmark, averaged over three data-generating seeds. The exact GP is included as a reference; it is not competitive on cost.

Method	RMSE(f)	coverage-95	$\hat{R}$	divergences	runtime (s)
HSGP	0.059	0.946	1.004	~ 1	309
DeepRV	0.075	0.857	1.009	~ 2	215
exact GP	0.060	0.959	1.005	58	1,692

The posterior-mean correlation of 0.997 between DeepRV and HSGP, which means the main structure of the field is recovered similarly, but DeepRV has speed advantage compared with HSGP.

The RMSE of Exact GP is 0.060, compared with 0.059 for HSGP, indicating little improvement in latent-field recovery in this setting. The interaction lengthscale is $\ell_{ws} = 0.15$, which is well represented by the eight HSGP basis functions used per spatial dimension. The exact GP is considerably more expensive and produces 58 divergent transitions.

We also varied the HSGP interaction basis from 768 to 2,160 and 5,120 functions. RMSE and coverage remain almost unchanged for these settings, but its runtime increases substantially (Appendix 3.1). This implies that the original basis already provides sufficient ability for this synthetic setting, and that increasing the number of basis functions offers little additional benefit.

Table 8: Variance decomposition across the three synthetic seeds. Values are percentages of total latent variance attributed to the spatial main effect g , temporal main effect q , and interaction w .

Seed	Truth ($g:q:w$)	HSGP ($g:q:w$)	DeepRV ($g:q:w$)
1	27:21:52	22:19:59	29:21:50
2	42:2:56	31:1:68	46:2:52
3	54:22:24	46:25:29	55:23:23

For each seed, the variance contribution of each latent component is calculated as its empirical variance divided by the sum of the empirical variances of g , q , and w , where the data has been centred to ensure no covariance exist. Overall speaking, both methods gives close results to the true variance decomposition. However, there is evidence that HSGP tends to overestimate the interaction term.

5.4 ECDC: an interaction at the resolution limit

We report the inverse-gamma specification of Table 2 throughout this section. Every fit under it converged, which is what lets the held-out comparison below carry weight. The bounded- ℓ_g variant arrives at is a sensitivity check, and Appendix 2 reports it.

On European surveillance data the picture changes from the birth series. The fitted interaction is short in space ($\hat{\ell}_{ws} = 0.078$ under HSGP) and carries most of the latent variance. Under HSGP this produces a lengthscale funnel: 34 divergent transitions and 1,569 seconds of sampling. DeepRV, whose decoder takes $\log \ell$ as an input and so never has to construct a near-resolution-limit field from a truncated basis, produces one divergence in 403 seconds (Table 9). In sample the two are close, with posterior-predictive coverage 0.985 against 0.988 and a WAIC difference of 17 ± 26 , or 0.6 standard errors, which shows little difference (Figure 15, appendix).

Held-out cross-validation. We take the cross-validation of §4.5 as the primary evidence. On 518 cells that never entered any likelihood, DeepRV and HSGP are tied on predictive density. The paired difference is 8.1 ± 13.5 , or 0.6 standard errors, with coverage 0.971 against 0.963 and runtimes of 152 against 622 seconds. Both fits converged ($\hat{R} = 1.012$ and 1.001). DeepRV attains HSGP’s predictive density on unseen data at roughly a quarter of the cost.

The differences are localised (Figure 6). HSGP’s fitted rate overshoots several outbreak peaks and produces excursions in periods where the observed counts are flat such as

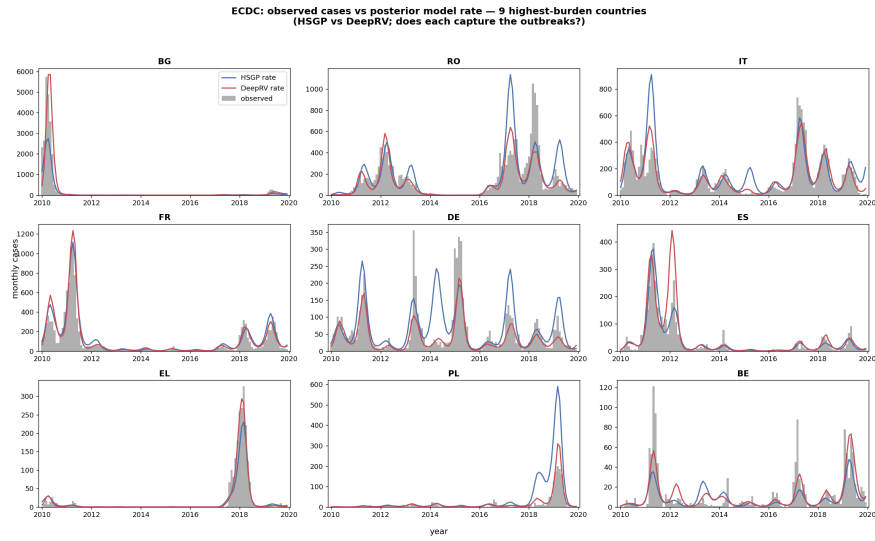


Figure 6: Posterior mean rate against observed counts for the nine highest-burden countries.

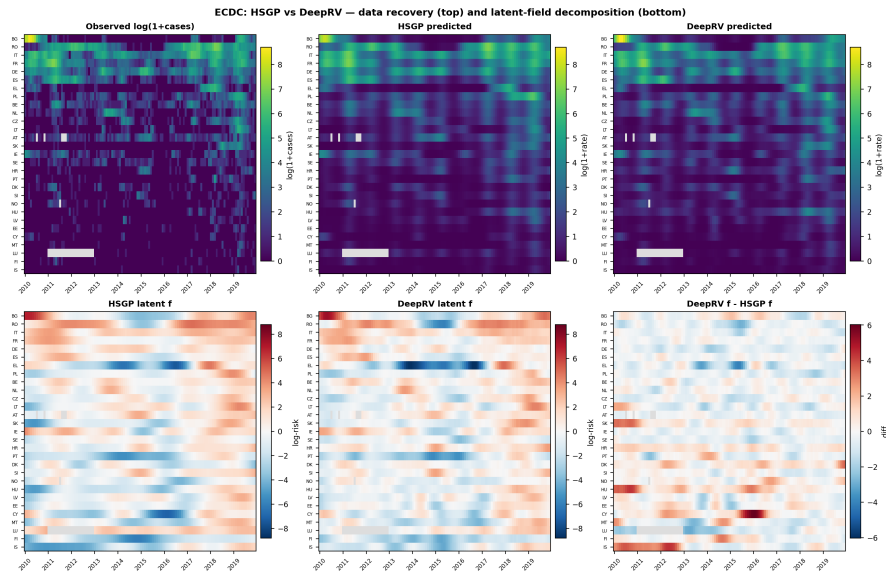


Figure 7: ECDC data recovery and latent-field decomposition, all 29 countries.

Table 9: ECDC measles, 2010–2019, four chains, under the inverse-gamma specification of Table 2.

	HSGP	DeepRV
<i>full fit</i>		
$\hat{R}$	1.005	1.006
divergences	34	1
runtime (s)	1,569	403
pp coverage-95	0.985	0.988
pp RMSE	86.0	86.8
ELPD (WAIC)	−7,018	−7,001
$\hat{\ell}_g$	0.53	3.72
var. share w	0.82	0.56
<i>held-out fit</i> , 518 withheld cells		
$\hat{R}$	1.001	1.012
ELPD	−1,110.2	−1,102.2
RMSE	52.3	32.9
coverage-95	0.963	0.971
runtime (s)	622	152
paired Δ ELPD	8.1 $\pm$ 13.5 (0.6 SE, tied)	

2014 in Romania and 2017 in Germany. DeepRV generally fits the data better. Calibration is adequate for both, with empirical coverage tracking nominal and near-uniform randomised PIT histograms (Figure 16, appendix).

Where the methods disagree. The interaction lengthscales differ, and the pattern is informative (Table 10; Figure 8 shows the full nine-parameter posteriors, and Table 18 in Appendix 3.2 the intervals). HSGP estimates $\hat{\ell}_{ws} = 0.078$, while DeepRV estimates $\hat{\ell}_{ws} = 0.104$. For HSGP, the posterior mass is concentrated near the lower resolution limit of the eight-function basis, so the estimate should be interpreted with caution. Figure 8 separates the two cases cleanly. The seasonal parameters, which never pass through either surrogate, coincide almost exactly, and so do the trend lengthscales; the components that differ are the ones the representations touch. This serves as an internal control, confirming that where the representations coincide the posteriors do too.

The variance decomposition differs too: HSGP attributes 82% of latent variance to the interaction against DeepRV’s 56%. Neither figure can be checked against a known answer on real data. The synthetic experiment, where truth is known and both methods converge, points to HSGP as the one over-attributing, with its decomposition error at 0.178 in L_1 against DeepRV’s 0.046. That indicates 82% is too high. The synthetic data already shows that HSGP tends to overweigh the interaction term.

ECDC: HSGP (blue) vs DeepRV (orange) 9-HP posterior densities

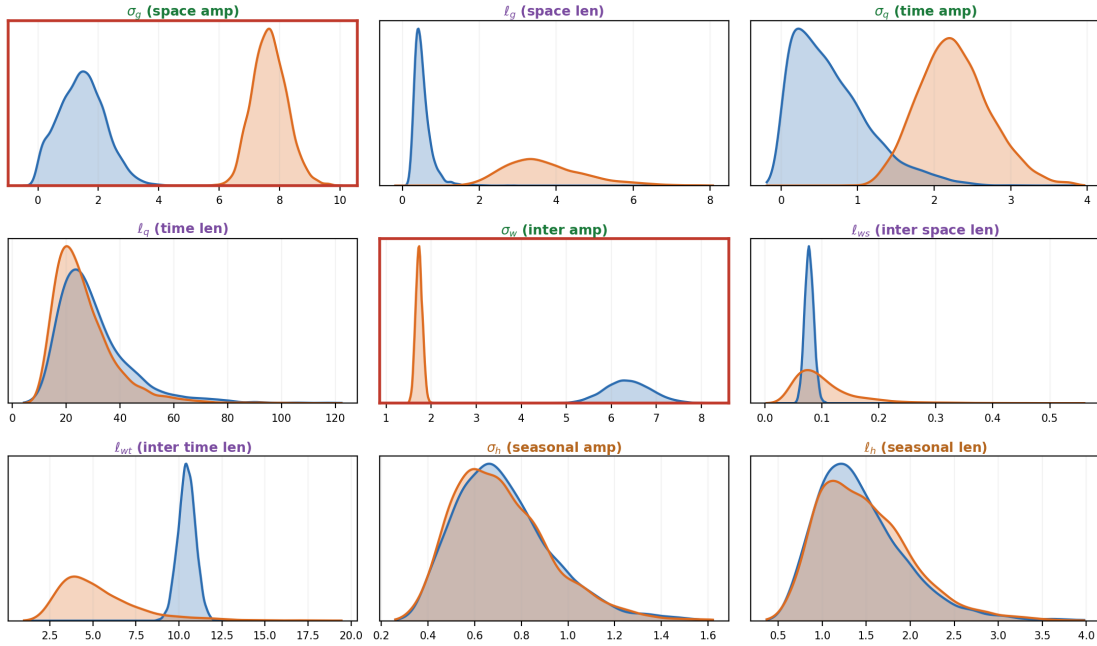


Figure 8: ECDC hyperparameter posteriors, HSGP (blue) against DeepRV (orange).

5.5 Monthly Tycho: the same comparison, six times the data

The pre-vaccine US panel is an order of magnitude larger than ECDC and behaves differently. The interaction term contribute less to the variance of the data compared to the formel two , and neither method funnels. After adapting the sampler (Appendix 1.3), both reach $\hat{R} = 1.007$ across all parameters with zero divergences (Table 11). DeepRV’s effective-sample-size distribution is 50.12% better than HSGP’s.

Their 95% Coverage achieved similar results. However, HSGP fits better in sample ($R^2 = 0.751$ against 0.674) and WAIC favours it by 33 standard errors. At this scale we regard that as a real difference. With a well-identified interaction and enough data to inform it, the complete linear basis has an advantage that the decoder’s finite capacity cannot match. DeepRV compensates in speed, running $2.7\times$ faster in wall-clock time from a 123-parameter latent space against roughly 900 basis weights.

Table 10: ECDC interaction and dispersion posteriors under the original specification. The seasonal and main-effect parameters, and the 95% intervals, are in Appendix 3.2. Components are compared through the variance decomposition.

Parameter	HSGP	DeepRV
ℓ_{ws} (interaction, space)	0.078	0.104
ℓ_{wt} (interaction, months)	10.5	5.4
α_w (interaction amplitude)	6.37	1.73
κ (over-dispersion)	0.84	0.97
$\text{var}(g):\text{var}(q):\text{var}(w)$	15:3:82	26:18:56

Table 11: Monthly Tycho, 1928–1962, four chains, $n = 19,871$.

Method	$\hat{R}$	min ESS	div.	runtime (s)	cov-95	R^2	$g:q:w$
HSGP	1.007	772	0	10,689	0.953	0.751	46:3:51
DeepRV	1.007	1,159	0	3,922	0.951	0.674	65:8:27

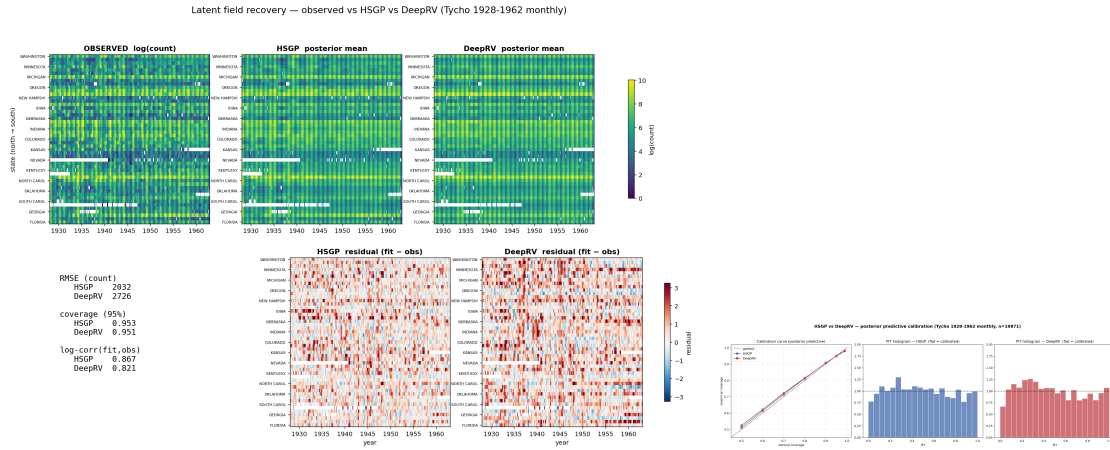


Figure 9: Monthly Tycho. Left: field recovery, with observed counts, both posterior means, and residuals. Right: predictive calibration; both curves sit on the diagonal and both PIT histograms are near uniform.

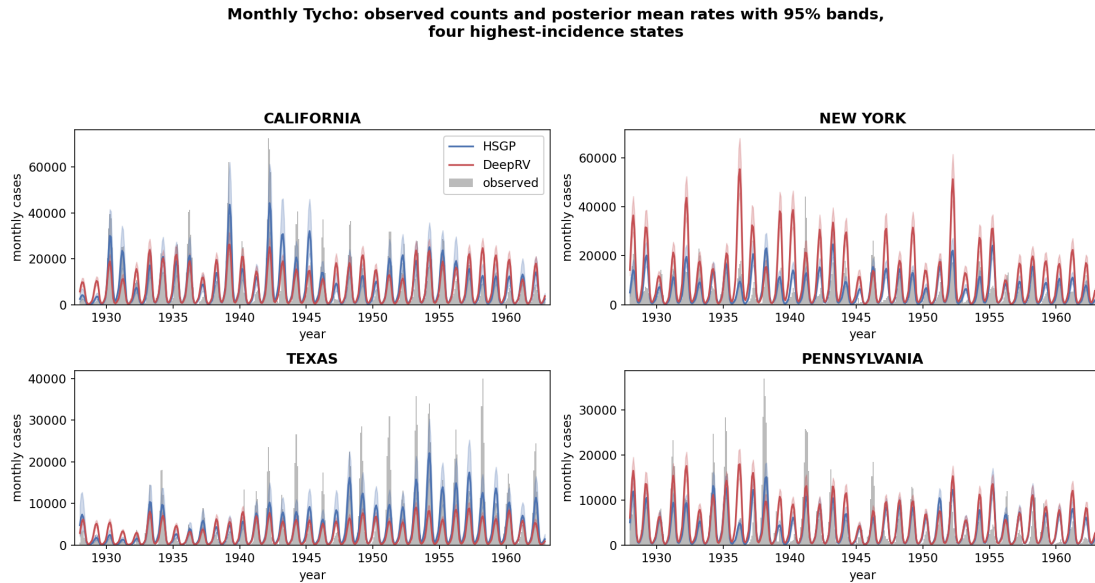


Figure 10: The four highest-incidence states: observed monthly counts with both posterior mean rates and 95% bands. Both methods reproduce the biennial epidemic rhythm and the timing of the peaks, but the amplitudes diverge by state and in opposite directions, with DeepRV running high in New York and low in Texas.

Figure 10 shows that both methods place the epidemics in the right months and reproduce the biennial alternation, but in terms of the amplitude of the peak, differences in prediction exist. For example, in New York DeepRV runs consistently above both HSGP and the data, reaching about 68,000 cases at the 1936 peak against an observed 40,000. In Texas it runs consistently below through the 1950s. Summed over the study period DeepRV assigns New York 1.85 times HSGP’s total and Wisconsin about half of it.

That being said, the correlation between the two posterior mean rate surfaces still has 0.83. Against the observed counts HSGP correlates at 0.81 and DeepRV at 0.66, with RMSE 2,010 against 2,703.

The methods agree on the season almost exactly and on the spatial main effect strongly (correlation 0.86). They diverge on the interaction (correlation 0.51), where DeepRV’s decoder renders a smoother field than HSGP’s basis (Figure 17, appendix). HSGP places more variance in the interaction (51% against 27%), which is consistent with our ECDC and synthetic data.

Figure 11 makes the difference concrete. HSGP’s interaction field is sharp and high-amplitude, with vertical bands marking epidemic years synchronised across many states.

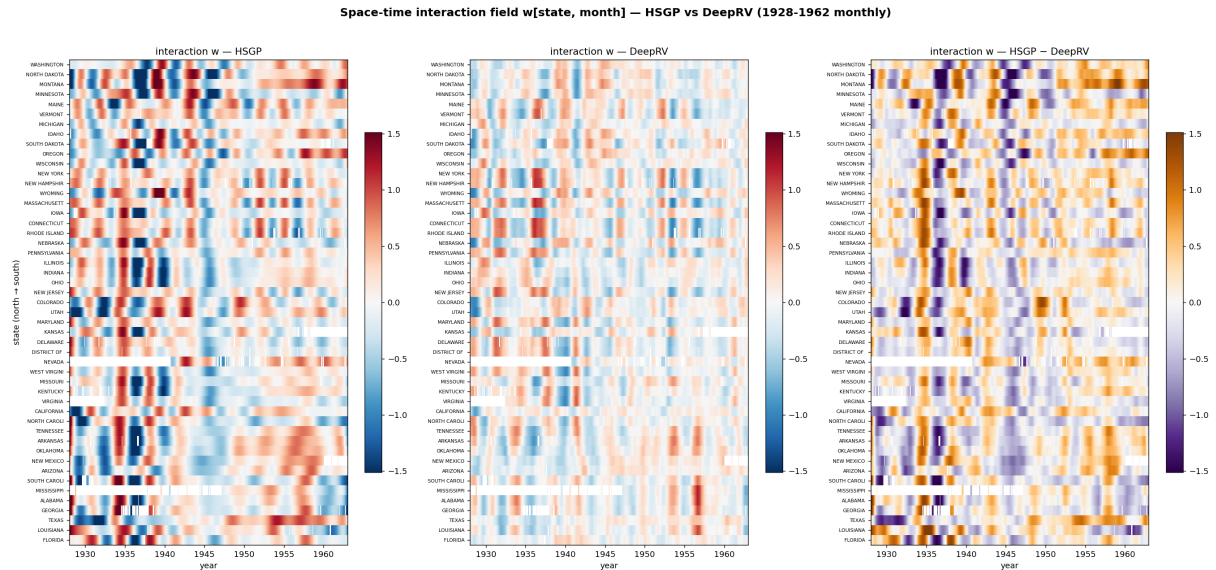


Figure 11: The space–time interaction field w on monthly Tycho. HSGP (left) renders a sharp, high-amplitude field; DeepRV (middle) recovers the same banding at reduced amplitude. The two agree on where the interaction is, and disagree on how much of it there is.

DeepRV recovers the same structure, with the bands appearing in the same places, but visibly less dense.

5.6 Importance of the interaction term

If the two representations differ only in how they render the space–time interaction, then removing that term should make them identical. We refitted every model with $f_{s,t} = g_s + q_t$ without the interaction term for both of the synthetic panel and ECDC.

Removing the interaction changes what the lengthscale priors should be. We had chosen the spatial prior to stay clear of the interaction-space prior, and with no interaction there is nothing to separate from. Keeping the old prior would handicap the ablated model twice over, once by removing a component and again by forbidding the remaining component from taking short lengthscales. We therefore re-derived both priors from pure containment over the full informative range. As it turns out this makes almost no difference to the fit, which is itself informative.

Three consequences follow (Table 12). The methods become indistinguishable: on the synthetic panel the HSGP and exact-GP fitted surfaces correlate at 0.99997, and on ECDC the three fitted surfaces correlate pairwise between 0.996 and 0.998. The exact

Table 12: Removing the interaction term. Synthetic errors are against the known latent field; ECDC errors are posterior-predictive on counts.

Dataset	Method	with interaction		without	
		RMSE	coverage	RMSE	coverage
Synthetic	HSGP	0.059	0.951	0.181	0.147
Synthetic	exact GP	0.060	0.959	0.181	0.147
ECDC	HSGP	86.0	0.985	170.3	0.974
ECDC	DeepRV	86.8	0.988	172.8	0.974
ECDC	exact GP	27.7	0.999	170.9	0.976

GP’s large ECDC advantage (RMSE 27.7 against 86.0) vanishes completely, and it ends up marginally behind HSGP (170.9 against 170.3). And every fit gets substantially worse.

The reason is visible in Figure 12. Without an interaction, the fit $f_{s,t} = g_s + q_t$ gives every country the same temporal profile up to a country-specific constant, so the shared trend q_t has to average over all of Europe. The model imposes the large 2019 outbreaks on countries even when there is no outbreak for them, and flattens Bulgaria’s 2010 epidemic to a few hundred cases.

We see the same effect in the over-dispersion parameter. On ECDC κ falls from 0.84–0.97 with the interaction to 0.29–0.31 without it. The variance still exist, which is re-attributed to the overdispersion term.

We report this as a sensitivity analysis rather than a candidate model. Dropping the interaction is not an option for measles. It does establish the claim the rest of this work rests on. The interaction is the only component on which these methods differ. A comparison of HSGP against DeepRV is therefore a comparison of two ways of representing w .

6 Discussion

6.1 What the evidence supports

For the ECDC data, DeepRV provides a practical alternative to HSGP, but we should interpret this conclusion with some caution. The comparison uses an HSGP interaction basis of $8 \times 8 \times 15$, and Table 3 indicates that it is too small to fully resolve the fitted lengthscale. However, the required number of basis functions is not computationally feasible here, as the Riutort-Mayol criterion gives $22^2 \times 13 \approx 6,300$ basis functions, which

is even more than the number of observations in the field. The benchmark therefore shows that DeepRV performs similarly to an HSGP when it is close to its resolution limit. However, it does not prove that this would always hold for a fully resolved HSGP, since we cannot fit such a model at comparable cost. Even so, this finding is still relevant, because the rapid tensor-product growth of the interaction basis would cause a severe computational burden for HSGP.

The held-out experiment is regarded as the most important evidence, as it does not rely on either an approximation benchmark or an in-sample fit statistic. For those observations not included in the likelihood, HSGP and DeepRV achieve predictive densities that agree within one standard error and show similar calibration, while DeepRV is four to six times faster.

The Tycho data show that there could be some disadvantage in point accuracy for DeepRV compared with HSGP. Both methods recover the epidemic cycles and achieve similar calibration, but they differ clearly in regional amplitude and interaction strength. HSGP provides the better in-sample fit, with WAIC favouring it by 33 standard errors, but DeepRV remains substantially faster. These results suggest that DeepRV’s performance relative to HSGP varies across settings. It achieves similar predictive density when the interaction is close to the resolution limit of the HSGP basis, but gives a poorer in-sample fit on Tycho.

The other costs are real but bounded. On the sparse synthetic data, DeepRV produces narrower credible intervals, with 0.86 coverage compared with 0.95 for HSGP. This undercoverage may reflect the finite capacity of the decoder, and makes DeepRV less suitable when reliable uncertainty quantification is a primary objective. The method also requires offline training, and its latent dimension is selected heuristically rather than by a formal criterion. Despite these limitations, DeepRV has a practical advantage on ECDC, where it sampled without substantial divergence.

6.2 Reconciling the ECDC and Tycho results

Although there seems to be some inconsistency, DeepRV represents the interaction terms adequately, and we regard the difference lies on HSGP’s side. On ECDC, the interaction-space containment prior is appropriate for the geometry, but the fitted interaction lengthscale is pushed down to 0.078, which is close to the smallest scale that an eight-function basis can represent. As a result, the lengthscale becomes difficult to identify and sampling becomes less stable. For Tycho data, the interaction is weaker and the fitted lengthscale stays well within the range that the basis can represent, so HSGP converges more cleanly.

Therefore, DeepRV’s advantage on ECDC does not mean that it is generally better than HSGP. Instead, it performs better in this case because it is less affected when the interaction lengthscale is closer to the resolution limit of the HSGP basis.

6.3 Implications for method comparisons

A fair comparison requires the fitted models to be aligned as closely as possible. During model checking, we identified differences in the kernel specification and seasonal parameterisation between two fits that were intended to represent the same model, and these were corrected before comparison. The choice of evaluation metric is also important. In-sample fit statistics can favour more flexible models, while leave-one-out approximations may become less reliable in such settings. We therefore place greater weight on held-out prediction, which is more computationally demanding but provides a more direct assessment of predictive performance.

6.4 Limitations

All three datasets for comparison are at monthly resolution. For finer resolutions, the interaction may become more complex, and DeepRV’s decoder has not been evaluated in such a setting. Further testing on daily and weekly data would be needed before extending the conclusions more generally.

The separable squared-exponential interaction is a modelling simplification. Measles spreads in travelling waves, and a non-separable kernel would represent this better. Whether it would raise or lower the effective rank, and hence which method it favours, remains an open question in our current setting.

Tycho uses the same population offset for all years, which is not fully accurate but affects HSGP and DeepRV in the same way. The exact GP comparison is only available for the two smaller panels. For the birth models, as Figure 1 shows, there are large single-day holiday effects that are not represented by any term in (10), so these remain in the residuals. Since both methods use the same model, this limitation affects them equally and does not change the comparison, but a special-day effect could be added to the model.

There is also a limitation specific to the DeepRV surrogate. The latent dimension is chosen using prior PCA fidelity, but there is no analytic rule for this choice. Unlike HSGP, which has the guideline in (5), DeepRV relies on a practical convention for latent sizing rather than a formal guarantee.

6.5 Future work

Several directions follow from discussion part directly. One is to test DeepRV at finer temporal resolutions and identify the point at which the decoder can no longer represent the interaction adequately. Because the decoder has finite capacity, this limit should depend on the effective rank of the interaction. Estimating this relationship could provide a practical rule for deciding when DeepRV is likely to remain reliable. Both the interaction rank and the resolution limit of a candidate HSGP basis can be examined using prior draws before fitting the model, which could help guide the choice of approximation in advance.

Another issue is the weak identifiability between a very short-range interaction and over-dispersion. In our results, the exact GP and the approximate methods separate these two sources of variation differently. A targeted synthetic experiment with controlled interaction lengthscales would be useful for studying this behaviour more directly.

7 Conclusion

Following our initial goal, we set out to test whether a neural surrogate prior can take the place of the Hilbert-space approximation, and the answer concluded from the five settings support is that establishment of this assumption is conditional rather than general.

Latent fields and their predictive consequences are largely preserved in all settings. Posterior mean trends on the two birth series are highly correlated with HSGP's at 0.9998 and 0.9999. On the held-out ECDC cells, HSGP and DeepRV achieve similar predictive density within one standard error and show comparable calibration, while DeepRV requires four to six times less computation, but the agreement is not as strong for hyperparameters and uncertainty estimates. On the Dutch birth series, the estimated trend lengthscale differs by about three pooled posterior standard deviations, but parameters that do not pass through the decoder are not affected. For the synthetic panel, DeepRV's 95% intervals achieve 85.7% coverage, which is lower than that of 94.6% for HSGP.

The contributing factor deciding which method performs better is probably the complexity of the space-time interaction. When the lengthscale approaches the resolution floor of the basis, as on ECDC, HSGP funnels and the surrogate is substantially faster at equal predictive density. When the interaction is well identified and sub-dominant in explaining the model's variance, as on Tycho, the complete linear basis fits better in sample. If the interaction term is removed and both of the methods produce very similar results close to Exact GP with a correlation of 0.9996 at the cost of poor point accuracy.

Here a practical recommendation follows from above. DeepRV is a suitable choice when computational cost is the main concern, when the interaction is close to the smallest scale that the HSGP basis can represent and when the main interest lies in the latent field or its predictive performance. HSGP is more suitable when the interaction can be represented adequately by the basis, when in-sample fit is the main priority. More generally, a surrogate prior should be evaluated by whether it preserves the underlying inference, not only by whether it reproduces similar fitted curves. Our results show that the latter can hold even when the former does not.

1 Implementation detail

1.1 Decoder architecture and training budget

All decoders share the architecture: two hidden layers of width 512 with tanh activations, taking $[\mathbf{z}, \log \ell]$ as input and returning the field. Training minimises mean squared reconstruction error with AdamW, learning rate 10^{-3} , weight decay 10^{-5} , batch size 128, for 6,000 steps on 6,000 exact GP fields per component. Fidelity is evaluated on 1,000 independently drawn fields. Offline training on CPU takes 6 seconds for the ECDC spatial decoder, 7 for the trend and 42 for the interaction.

1.2 Numerical stabilisation of the exact GP

The Cholesky jitter matters more than one might expect. At 10^{-5} the very smooth main-effect kernels have condition numbers around 4×10^6 and individual chains collapse to step sizes of 10^{-5} , while 10^{-4} brings the condition number to 10^5 and the sampler behaves. The failure is chain-specific, so a single-chain trial run does not catch it.

1.3 Getting the monthly Tycho model to converge

The monthly Tycho HSGP fit, with roughly 900 interaction basis weights and a tight pre-vaccine likelihood, initially failed to mix, at $\hat{R}$ of 16. Convergence is a computational problem and we fixed it computationally, without deleting basis functions to flatter the diagnostic (Table 19). Initialising at the hyperparameter modes puts every chain in the typical set from the first iteration; a dense mass matrix on the hyperparameter block handles the strong amplitude-lengthscale correlation that a diagonal metric cannot; and four chains with 2,000 post-warm-up draws gave enough resolution. The final fit has

$\hat{R} \leq 1.007$ across all parameters, including every basis weight, with minimum effective sample size above 770.

2 The spatial-lengthscale refit sequence

This appendix records the sequence of attempted fixes for ECDC data, since the failures are what establish that the problem is identifiability rather than decoder coverage.

The audit found 18.7% of the posterior mass for ℓ_g beyond the largest lengthscale in the decoder’s training sample, which spans $[0.18, 4.60]$. *Retraining the spatial decoder* on $\ell_g \in [0.08, 12]$ removed the extrapolation and improved reconstruction fidelity substantially where it mattered: measured on fields drawn at the fitted posterior lengthscales, ρ^2 rose from 0.85 to 0.998 and RMSE fell by a factor of three. The refitted model nonetheless failed to converge, with $\hat{R} = 7.4$, minimum ESS of 2, and a posterior mean for ℓ_g of 3.1×10^4 . *Capping the training range* at $[0.03, 2.0]$ did not help either, because the training range constrains the decoder and not the sampler: NUTS proposes whatever the prior allows and the network extrapolates. That version drifted to $\hat{\ell}_g = 69$.

Re-deriving the containment prior over the informative band $[0.030, 1.331]$, which is what Betancourt’s argument prescribes and what the original prior failed to do since its 98% upper limit of 1.80 sits above the domain span of 1.33, gives $\text{InvGamma}(2.007, 0.200)$. This also fails, and instructively. Narrowing the band forces the shape parameter down from 6.11 to 2.01, which makes the tail heavier: the density now decays as $\ell^{-3.01}$ rather than $\ell^{-7.11}$, and the prior mass above $\ell = 10$ rises from 10^{-6} to 1.9×10^{-4} . A containment prior controls where the central 98% of the mass sits, not how fast the tail decays, and when the likelihood is flat it is the tail that decides. The refitted model drifted to $\hat{\ell}_g = 31$ with a 95% interval of $[0.01, 213]$. Only bounding the support, with $\ell_g \sim \text{LogUniform}(0.030, 1.331)$, contains the parameter.

Table 13: Sensitivity of the ECDC DeepRV fit to the spatial-decoder configuration.

Across four treatments spanning four orders of magnitude in $\hat{\ell}_g$, the predictive quantities are stable while the variance decomposition shifts.

Spatial treatment	$\hat{\ell}_g$	$\hat{\alpha}_g$	pp cov-95	pp RMSE	held-out ELPD	var. share w
original (prior-trained)	3.7	7.68	0.988	86.8	−1102.2	0.556
decoder widened $[0.08, 12]$	3.1×10^4	1.48	0.987	85.2	−1096.1	0.472
decoder capped $[0.03, 2]$	69.5	1.66	0.989	78.6	−1068.2	0.488
prior widened, band $[0.03, 1.33]$	31.2	1.63	0.990	78.6	−1104.3	0.460

Table 14: Decoder retraining at three seeds, varying both the sampled training fields and the network initialisation. Training is deterministic given a seed: rerunning the original configuration reproduced the stored decoder metrics exactly.

Decoder	prior RMSE			fidelity ρ^2		
	seed 1	seed 2	seed 3	seed 1	seed 2	seed 3
spatial g	0.0388	0.0408	0.0447	0.969	0.953	0.956
trend q	0.0403	0.0465	0.0406	1.002	0.995	0.985
interaction w	0.3544	0.3547	0.3583	0.847	0.846	0.844

3 Supplementary tables and figures

3.1 Synthetic benchmark

Table 16 gives the per-seed synthetic results. Table 15 gives the HSGP interaction-basis sweep referred to in §5.3: increasing the basis sevenfold leaves recovery error and coverage unchanged while nearly tripling runtime.

Table 15: HSGP interaction-basis sweep on the synthetic panel, seed 1.

Interaction basis functions	768	2,160	5,120
RMSE(f)	0.0594	0.0605	0.0604
coverage-95	0.951	0.961	0.961
runtime (s)	309	520	874

Table 16: Synthetic benchmark, per seed. Errors are against the known latent field.

Method	seed	RMSE(f)	coverage	runtime (s)	div.
HSGP	1	0.0594	0.950	309	0
HSGP	2	0.0555	0.954	269	2
HSGP	3	0.0620	0.933	348	2
DeepRV	1	0.0759	0.862	266	0
DeepRV	2	0.0701	0.869	146	0
DeepRV	3	0.0797	0.840	233	0

Table 17: ECDC held-out cross-validation. A shared random 15% of cells (518 of 3,450) was withheld from the likelihood; all models were refitted on the remainder.

Method	held-out ELPD	RMSE	cov-95	$\hat{R}$	div.	runtime (s)
exact GP	-965.1 ± 45.1	21.9	0.971	1.154	18	7,196
DeepRV	$-1,102.2 \pm 50.6$	32.9	0.971	1.012	3	152
HSGP	$-1,110.2 \pm 50.5$	52.3	0.963	1.001	7	622

Paired differences: DeepRV – HSGP = 8.1 ± 13.5 (0.6 SE, tied); exact GP – DeepRV = 137.1 ± 18.9 (7.3 SE).

3.2 Full ECDC hyperparameter posteriors

Table 18: ECDC hyperparameter posteriors under the original specification, mean with 95% interval. Seasonal parameters agree across methods because the season bypasses both surrogates; the interaction lengthscales differ because of what each representation can express. The amplitudes are not on a common scale, since HSGP’s multiply a basis expansion while DeepRV’s and the exact GP’s multiply a variance-normalised field, which is why components are compared through the variance decomposition.

Parameter	HSGP	exact GP	DeepRV
α_g (spatial amplitude)	1.46 [0.12, 2.99]	2.16 [1.47, 2.99]	7.68 [6.66, 8.85]
ℓ_g (spatial)	0.53 [0.26, 1.11]	0.11 [0.08, 0.13]	3.72 [2.11, 6.20]
α_q (trend amplitude)	0.66 [0.03, 1.90]	1.19 [0.65, 2.08]	2.26 [1.44, 3.24]
ℓ_q (trend, months)	30.3 [13.6, 68.6]	17.6 [11.1, 27.0]	26.5 [12.5, 55.8]
α_w (interaction amplitude)	6.37 [5.42, 7.41]	1.73 [1.62, 1.84]	1.73 [1.59, 1.88]
ℓ_{ws} (interaction, space)	0.078	0.023	0.104
ℓ_{wt} (interaction, months)	10.5	2.0	5.4
σ_h (seasonal amplitude)	0.73	0.68	0.72
ℓ_h (seasonal)	1.45	1.07	1.49
κ (over-dispersion)	0.84	12.89	0.97
$\text{var}(g):\text{var}(q):\text{var}(w)$	15:3:82	27:14:59	26:18:56

3.3 Monthly Tycho convergence

Table 19: Convergence-fixing sequence for the monthly Tycho HSGP fit. No basis functions were removed.

Configuration	$\hat{R}$	note
diagonal metric, no initialisation, 600 warm-up	16.3	chains move but do not mix
+ initialise at hyperparameter modes	3.1	starts in the typical set
+ dense mass matrix on the hyperparameter block	1.02	handles α - ℓ correlation
+ four chains, 2,000 post-warm-up draws	1.007	all parameters; min ESS 772

3.4 Supplementary figures

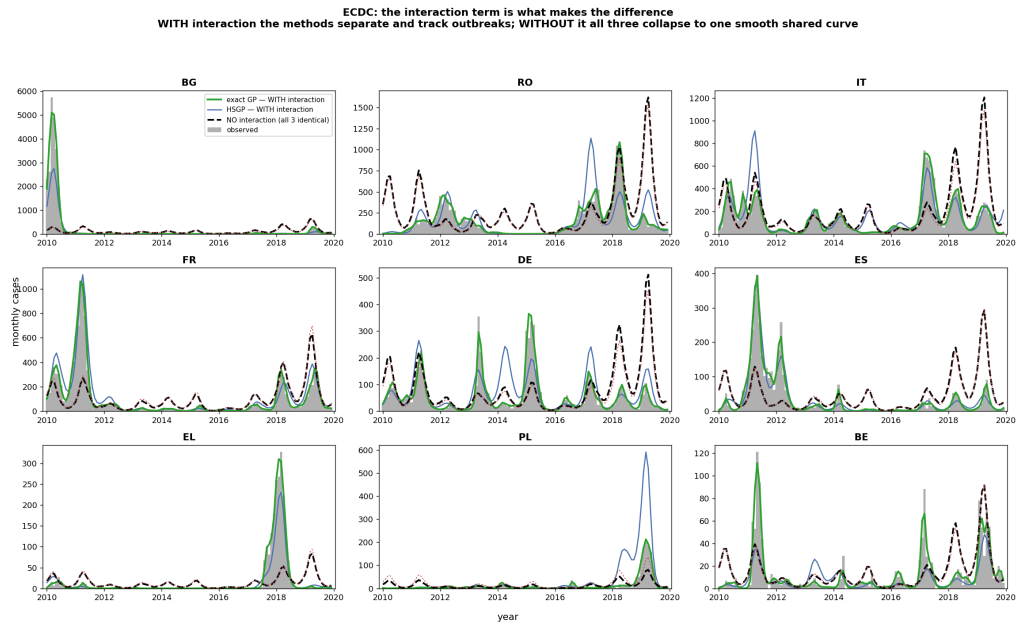


Figure 12: ECDC with and without the interaction term. No-interaction fit

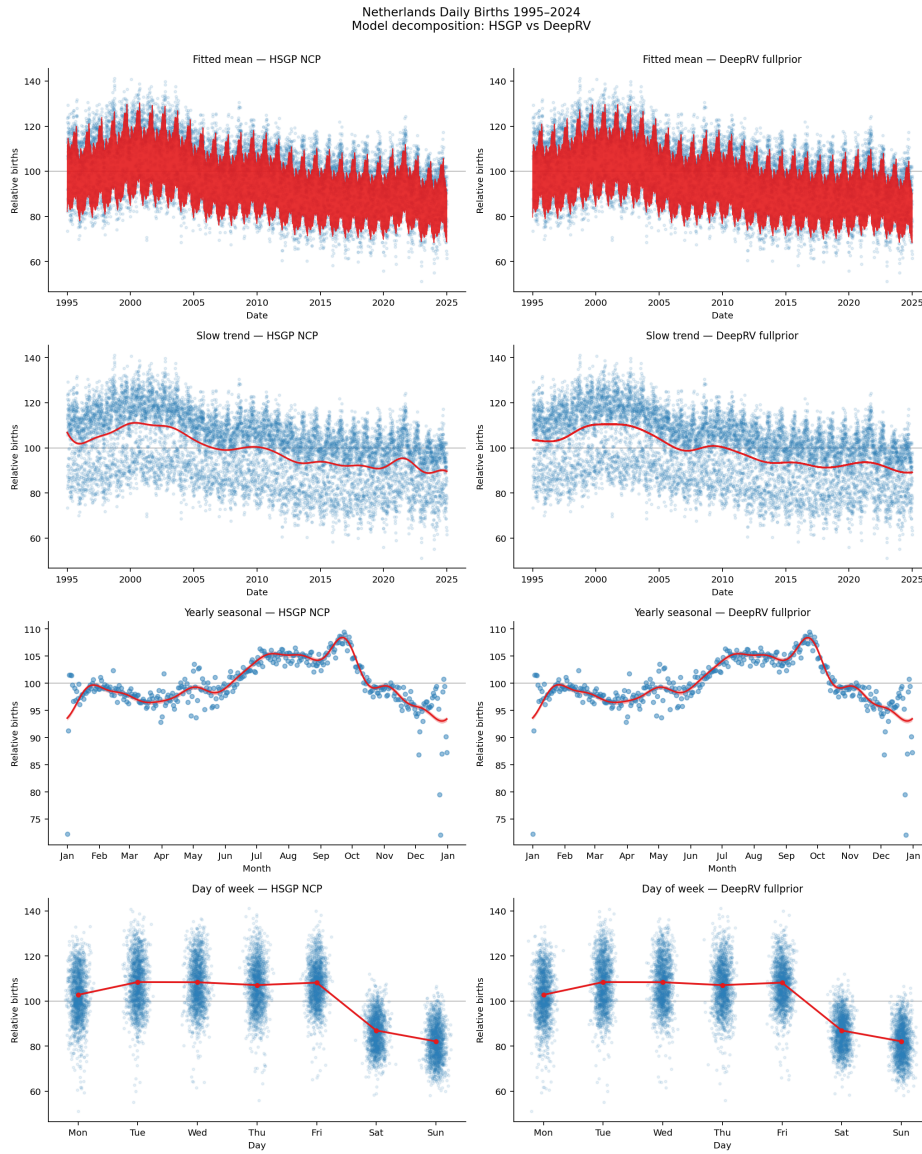


Figure 13: Netherlands births: fitted decomposition under HSGP (left) and full-prior DeepRV (right). Rows: fitted mean, slow trend, annual cycle, weekday effect. Correlation between the posterior mean linear predictors is 0.9999.

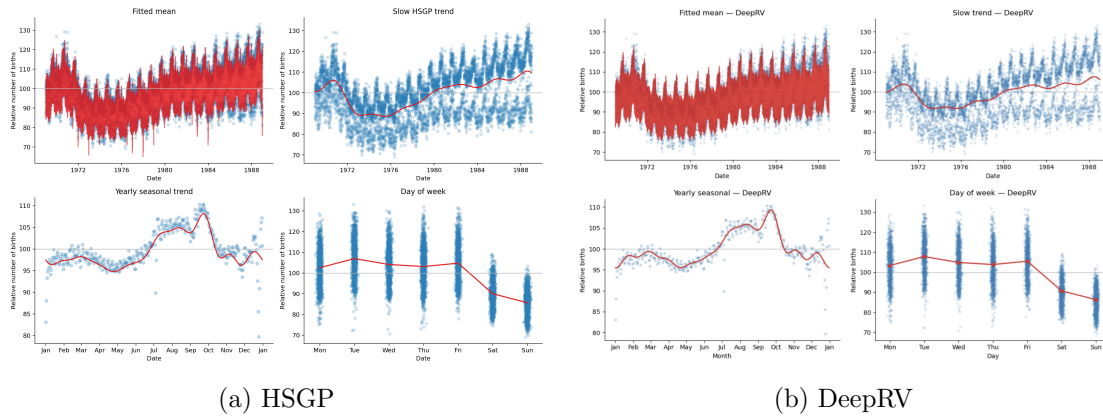


Figure 14: US daily births: fitted decomposition under the two representations. Panels within each block: fitted mean, slow trend, annual cycle, weekday effect. The two blocks are difficult to tell apart, and the posterior mean trends correlate at 0.9998. Several points near the year boundary sit far below the fitted seasonal curve under both methods, these being the holiday effects the model has no component for.

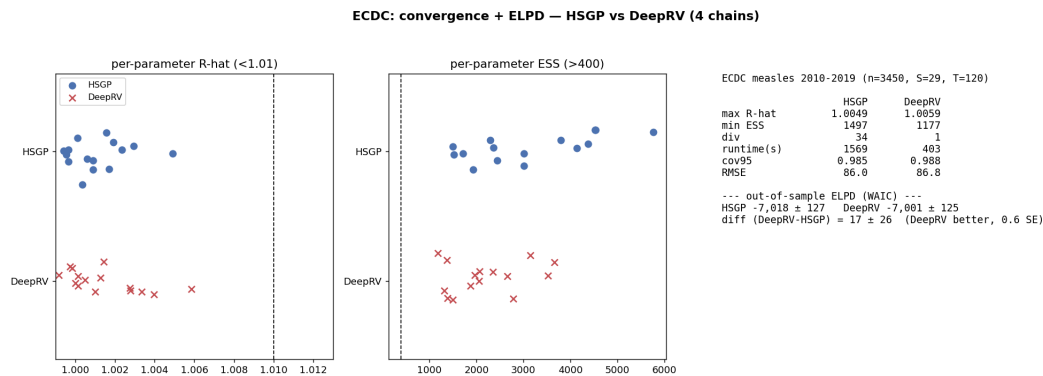


Figure 15: ECDC convergence and predictive density under the original specification. Every parameter satisfies $\hat{R} < 1.006$ with effective sample size above 1,000 for both methods, and the WAIC difference is well within one standard error. The difference that matters is the divergence count and the runtime.

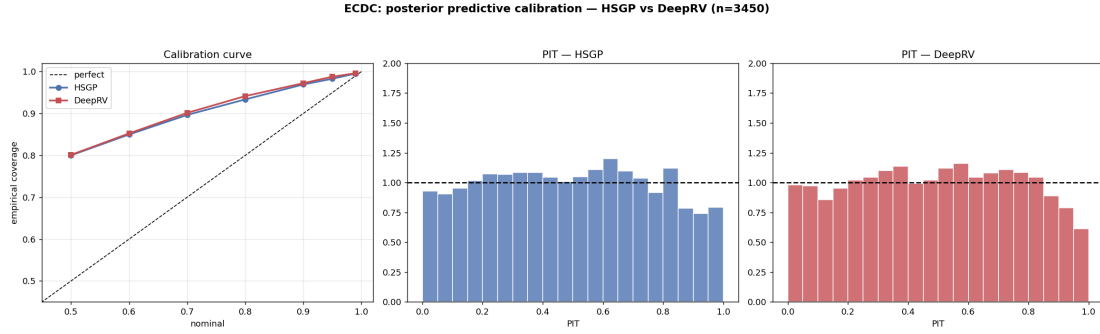


Figure 16: Posterior-predictive calibration on ECDC. Empirical coverage tracks nominal for both methods, and both randomised PIT histograms are close to uniform. Neither method is overconfident, which is the check the VAE-based predecessors of DeepRV fail.

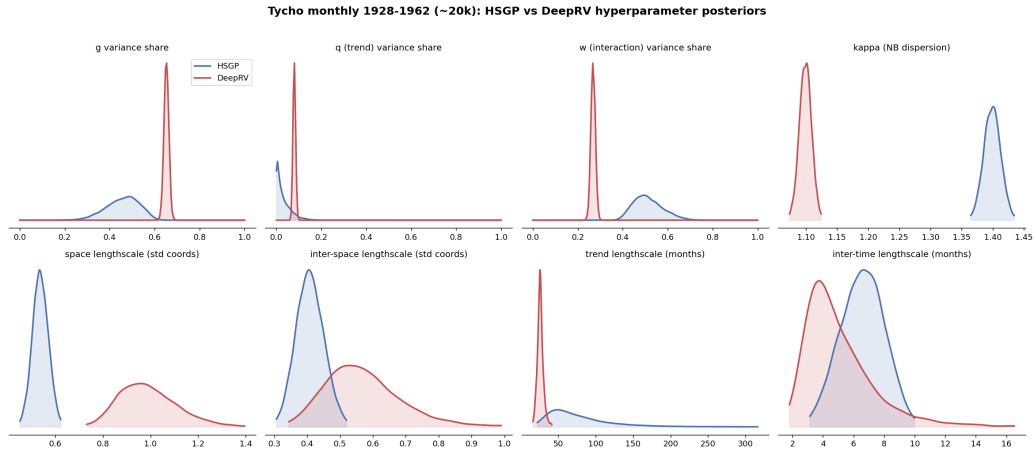


Figure 17: Hyperparameter posteriors, monthly Tycho. The shared kernel hyperparameters overlap closely. The amplitude parameters differ in scale by construction, since HSGP’s multiplies a basis expansion and DeepRV’s a variance-normalised decoder output, which is why components are compared through the variance decomposition instead.

4 Reproducibility

All models are implemented in NumPyro. Each dataset has a self-contained pipeline: preprocessing that assembles the count panel, exposure offset and standardised coordinates; decoder training on exact GP draws; and paired inference scripts that share kernel, seasonal component, priors and sampler settings so that only the latent representation differs. Every run stores its full posterior, comprising per-parameter $\hat{R}$ and effective sample size, chain-wise traces, pointwise log-likelihood, posterior-predictive draws and latent field draws, so that all figures and tables are regenerated from saved output without refitting.

Data availability. Measles counts are the ECDC confirmed-case monthly series (ECDC, 2026), downloaded 23 June 2026. Population denominators are Eurostat `demo_pjan` (population on 1 January, total), 2010–2019, retrieved through the Eurostat REST API on 11 July 2026 (Eurostat, 2026). US counts are from Project Tycho (van Panhuis et al., 2013). All three are publicly available. Synthetic generation, decoder training and inference seeds are fixed and recorded; inference uses NumPyro (Phan et al., 2019).

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